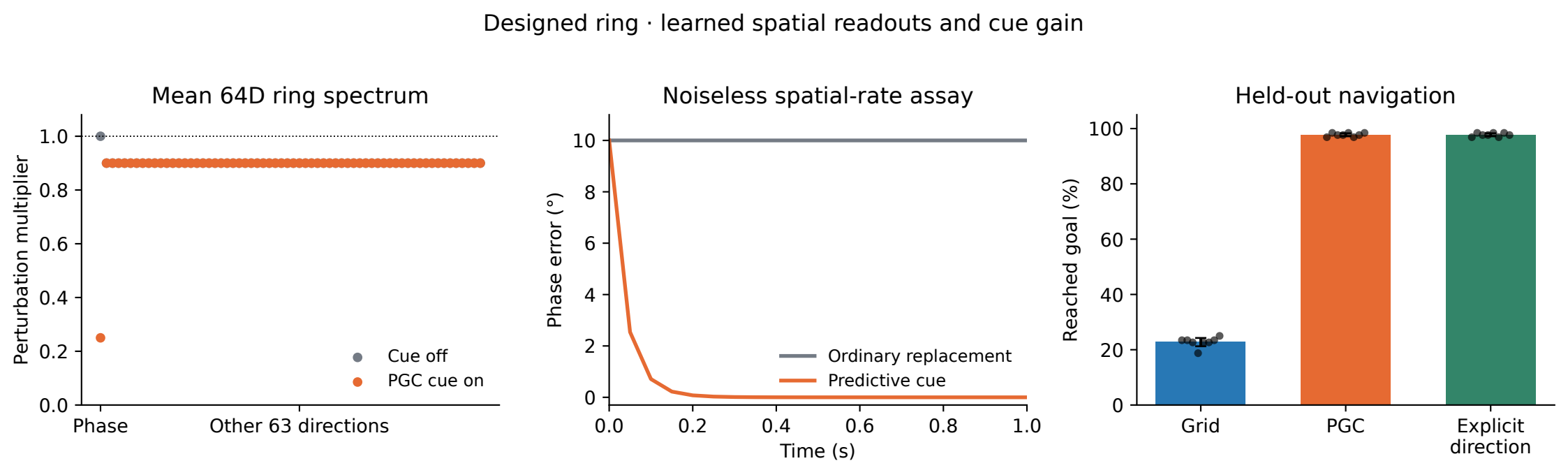
Designed ring · learned spatial readouts and cue gain
Mean 64D ring spectrum
Perturbation multiplier
1.0
0.8
0.6
0.4
0.2
0.0
Cue off
PGC cue on
Phase
Other 63 directions
Noiseless spatial-rate assay
Phase error (°)
10
8
6
4
2
0
Ordinary replacement
Predictive cue
0.0
0.2
0.4
0.6
0.8
1.0
Time (s)
Held-out navigation
Reached goal (%)
100
80
60
40
20
0
Grid
PGC
Explicit direction